

Jenga Hemant Rao

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SUMMARY

Frontend Engineer specializing in AI-native web applications. Experienced in building RAG pipelines, multi-agent AI systems, and production-grade UIs using Next.js, TypeScript, and LangChain JS. Proven ability to architect efficient AI systems under constrained environments — optimizing for performance, cost, and scalability. Seeking a Software Engineer / AI-Web Developer role to build intelligent, user-centric products at scale.

TECHNICAL SKILLS

Languages: JavaScript (ES2023+), TypeScript, HTML5, CSS3

Frameworks & Libraries: Next.js 14, React.js, Redux Toolkit, Tailwind CSS

AI & ML: LangChain JS, LangGraph JS, LangSmith JS, RAG Pipelines, Multi-Agent Systems

LLM Infrastructure: Ollama (local LLM), Groq LPU Inference, OpenAI API, HuggingFace

Tools & Platforms: Git, GitHub, Vercel,

PROJECTS

- Multi-Agent AI Task Orchestration System** | *LangChain JS, LangGraph, Groq, TypeScript* | [Live Demo](#) 2026
- Built and deployed an AI-powered product launch studio that transforms raw product ideas into positioning, target audience, launch messaging, strategy, and risk analysis.
 - Designed a LangGraph-based multi-agent workflow with specialized AI agents for research, product strategy, marketing, and final review.
 - Integrated Groq LPU inference and Ollama through a server-side Next.js API architecture, enabling secure and resource-aware LLM execution.
 - Developed and deployed the production-ready application on Vercel, with a responsive React/Next.js interface and server-side API integration.
- DocuMind AI** | *React.js, TypeScript, Tailwind CSS, LangChain.js, RAG, Groq API, PDF.js* | [Live Demo](#) 2026
- Engineered a production-style Retrieval-Augmented Generation (RAG) application that enables users to upload multiple PDF documents, build a searchable knowledge base, and receive grounded AI-generated answers with page-level citations.
 - Built a responsive document management system supporting multi-file uploads, document search, deletion, real-time indexing, and citation-aware chat for accurate document-based Q&A.
 - Developed a modern React + TypeScript frontend with Tailwind CSS, dark/light mode, streaming chat interface, and reusable UI components, then deployed the application on Vercel with a Node.js backend.
- Nexa AI Helpdesk** | *React.js, JavaScript, LangChain.js, LangGraph, Groq API* | [Live Demo](#) 2026
- Built and deployed an AI-powered customer support chatbot capable of handling account, billing, and technical support queries through a conversational chat interface.
 - Implemented conversational memory using LangGraph persistence, enabling the assistant to retain relevant user context across messages and provide context-aware responses.
 - Integrated a LangChain/LangGraph-based AI workflow with Groq API for LLM-powered response generation and structured conversation handling.
 - Developed and deployed an Express.js API service with health monitoring and CORS configuration, connecting the React frontend with the AI helpdesk workflow.
- Nexus Admin Dashboard** | *React.js, TypeScript, Tailwind CSS* | [Live Demo](#) 2026
- Architected a high-performance admin management panel featuring real-time data tables, role-based access control, and complex nested state management.
 - Implemented optimistic UI updates and request deduplication, reducing perceived latency by 60% and eliminating redundant API calls.
 - Built reusable, fully typed React component library with 20+ components, enforcing strict TypeScript interfaces for maintainability and scalability.

EDUCATION

University of Calcutta

Kolkata, India

Bachelor of Commerce — Syama Prasad College

2021 – 2026

- Self-directed specialization in AI/ML engineering, frontend development, and LLM systems architecture

Languages: English (Professional), Hindi (Native), Telugu (Native)